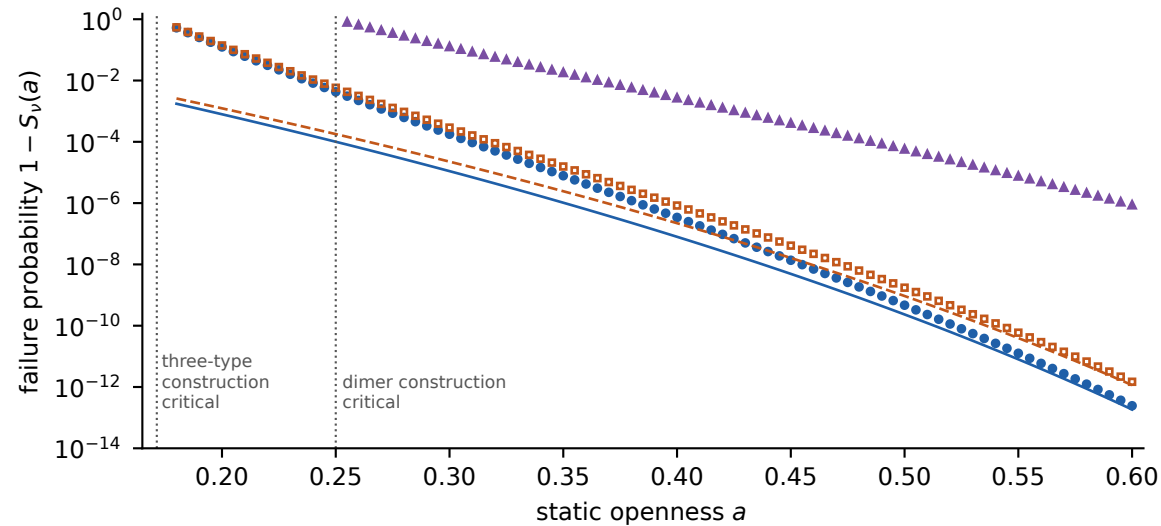




- three-type certificate, split
- ◻ three-type certificate, quotient
- ▲ dimer certificate (both)

- gateway lower bound $(1 - a)^{32}$
- - - gateway lower bound $(1 - a)^{30}$